NYC Yellow Taxi Trip Records

Insights and Trends

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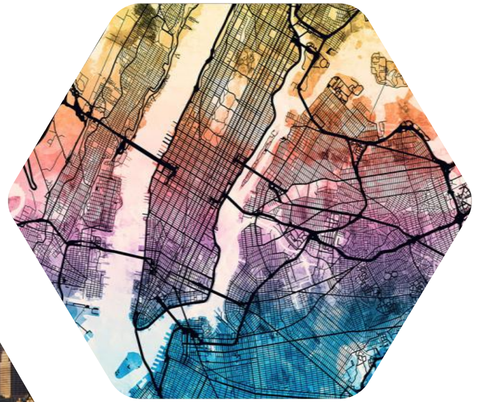
8 June 2026

OUTLINE

1. Introduction
2. Data quality
3. Data cleaning
4. Feature engineering
5. Data analysis
6. Conclusions



OUTLINE

1. Introduction
   - 1.1 Dataset
   - 1.2 Research questions
   - 1.3 Business significance
   - 1.4 Methods
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3. Data cleaning
4. Feature engineering
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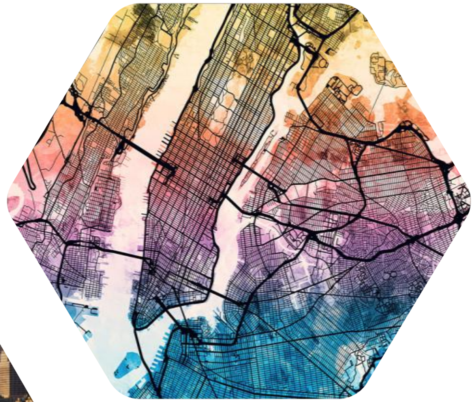


1. INTRODUCTION

1.1 Dataset

- **~305,000** NYC Yellow Taxi trips from **January 2018 to January 2023**
- 2018–2022: **~60,000** trips per year; 2023: January only (~**5,000** trips)
- Data appears sample-capped at ~5,000 trips/month - ratio metrics are reliable, absolute volumes are not.
- **19 columns**:
    - Pickup & dropoff time
    - Pickup & dropoff zone ID
    - Trip distance
    - Other dimensions (vendor, rate code, payment type, etc.)
    - Financial columns (fare, tax, surcharge, total, etc.)

```
RangeIndex: 304978 entries, 0 to 304977
Data columns (total 19 columns):
 #   Column                 Non-Null Count   Dtype
---  ------                 --------------   -----
 0   VendorID               304978 non-null  category
 1   tpep_pickup_datetime   304978 non-null  datetime64[ns]
 2   tpep_dropoff_datetime  304978 non-null  datetime64[ns]
 3   passenger_count        295465 non-null  category
 4   trip_distance          304978 non-null  float64
 5   RatecodeID             295465 non-null  category
 6   store_and_fwd_flag     295465 non-null  category
 7   PULocationID           304978 non-null  category
 8   DOLocationID           304978 non-null  category
 9   payment_type           304978 non-null  category
 10  fare_amount            304978 non-null  float64
 11  extra                  304978 non-null  float64
 12  mta_tax                304978 non-null  float64
 13  tip_amount             304978 non-null  float64
 14  tolls_amount           304978 non-null  float64
 15  improvement_surcharge  304978 non-null  float64
 16  total_amount           304978 non-null  float64
 17  congestion_surcharge   232346 non-null  float64
 18  airport_fee            106217 non-null  float64
dtypes: category(7), datetime64[ns](2), float64(10)
```

1. INTRODUCTION

1.2 Research Questions

Analyzed in section	Topics
5.1 Overview	Citywide demand, revenue, payment, and efficiency patterns
5.2 Time	Demand and efficiency by month, weekday, and hour
5.3 Space	Demand and efficiency hotspots by zone and borough
5.4 Vendor	Vendor performance comparison
5.5 Rate code	Rate-code segment performance comparison
5.6 Tip	Tip behavior by time and location
5.7 Fare	Relationship between fare, distance, and duration

1. INTRODUCTION

1.3 Business significance

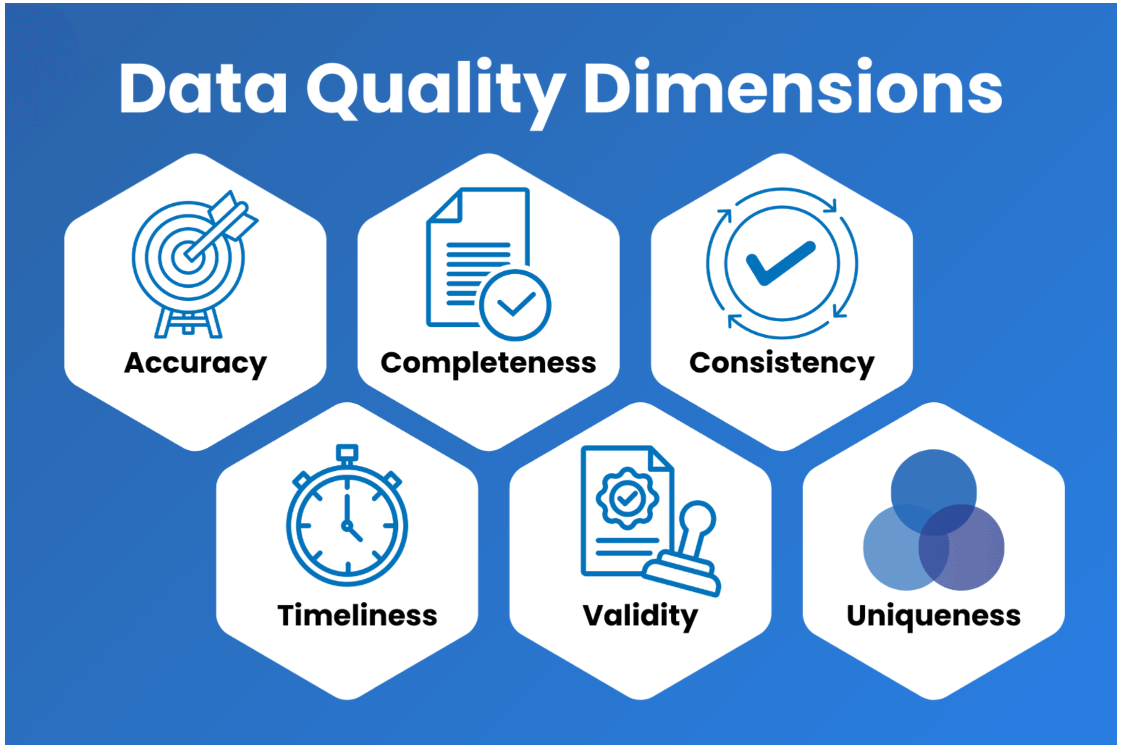
- Provide **clear, actionable KPIs**.
- Support **smarter fleet allocation.**
- Optimize **revenue management**.
- Enable more **targeted service strategy**.
- Improve **data quality**.

1. INTRODUCTION

1.4 Method

Data preparation → Data Quality Check → Data Cleaning → Feature Engineering → Analysis → Conclusions

2. DATA QUALITY



In the scope of this assignment, I will focus on Completeness, Uniqueness, Validity and Accuracy.

The other 2 dimensions are not relevant in this assignment:

- Timeliness (how up-to-date data are): this is a static dataset
- Consistency (data are consistent among tables, systems): there is only 1 dataset

2. DATA QUALITY

2.1 Completeness

- ~3% of rows had NULL values in passenger_count, RatecodeID, and store_and_fwd_flag — all three missing together, suggesting a systematic data collection issue
- congestion_surcharge and airport_fee have NULLs that are not substitutes for zero — retained as missing

2.2 Uniqueness

- 3 pairs of duplicate trips identified based on identical pickup/dropoff time and distance
- Each pair showed a specific issue: NULL metadata (2 pairs) or negative financial values (1 pair)

2.3 Validity

- VendorID contains undocumented values (4 and 5) - treated as “Others”
- Negative values present in all numeric columns (less than 1%); zero-distance trips peaked in 2020
- Store-and-forward trips exhibit a substantially higher rate of zero-distance records (4.41%) compared with non-store-and-forward trips (1.20%)

2.4 Accuracy

- Implausible trips found: >100 miles, >3 hours duration, >100 mph speed
- total_amount mismatches linked to congestion_surcharge recording errors (confirmed by chi-square test)

3. DATA CLEANING

3.1 Completeness: NULL imputation

- passenger_count, RatecodeID, store_and_fwd_flag: mode-imputed with a missingness flag column added
- congestion_surcharge and airport_fee: retained as NULL (not forced to 0)

3.2 Uniqueness: duplicate removal

- Deleted the NULL-metadata row from each of the 2 metadata duplicate pairs
- Deleted the negative-value row from the 1 financial duplicate pair

3.3 Validity: remove negative values

- Dropped rows with negative trip_distance and financial columns
- Zero values retained — removing them silently would undercount minimum-fare and cancellation revenue

3.4 Accuracy: plausibility filters

- Kept only trips with: distance ≤100 miles, duration ≤180 minutes, speed ≤100 mph
- total_amount mismatches not corrected - no reliable business rule to determine which field is wrong

Result

- 0.82% of rows removed; cleaned dataset retains over 302k trips with more reasonable distributions

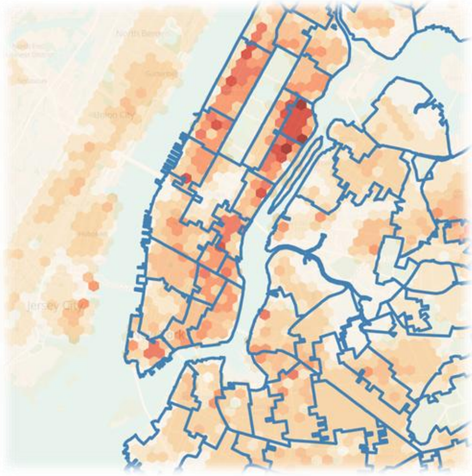
4. FEATURE ENGINEERING

4.1 New datetime and derived columns

- Pickup year, month, day-of-week, hour of day - for temporal analysis
- trip_duration_min, avg_speed_mph - for accuracy checks and efficiency analysis
- Human-readable labels for VendorID, RatecodeID, payment_type

4.2 Geospatial enrichment

- Joined TLC zone lookup table: added pickup/dropoff borough, zone name, and service zone
- Loaded TLC zone shapefiles for geographic heatmap visualizations



4.3 Sample Validation

Representativeness:

- Tested payment mix, borough share, and hourly distribution stability across months
- Sample composition is internally consistent; ratio metrics are suitable for trend analysis

COVID Impact:

- Many metrics experienced disruption in 2020
- All Correlation with COVID-19 noted; causality cannot be established from this sample alone

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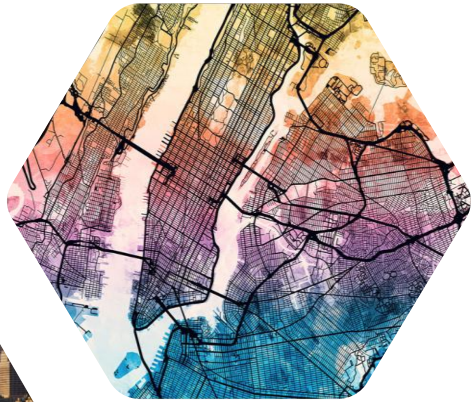
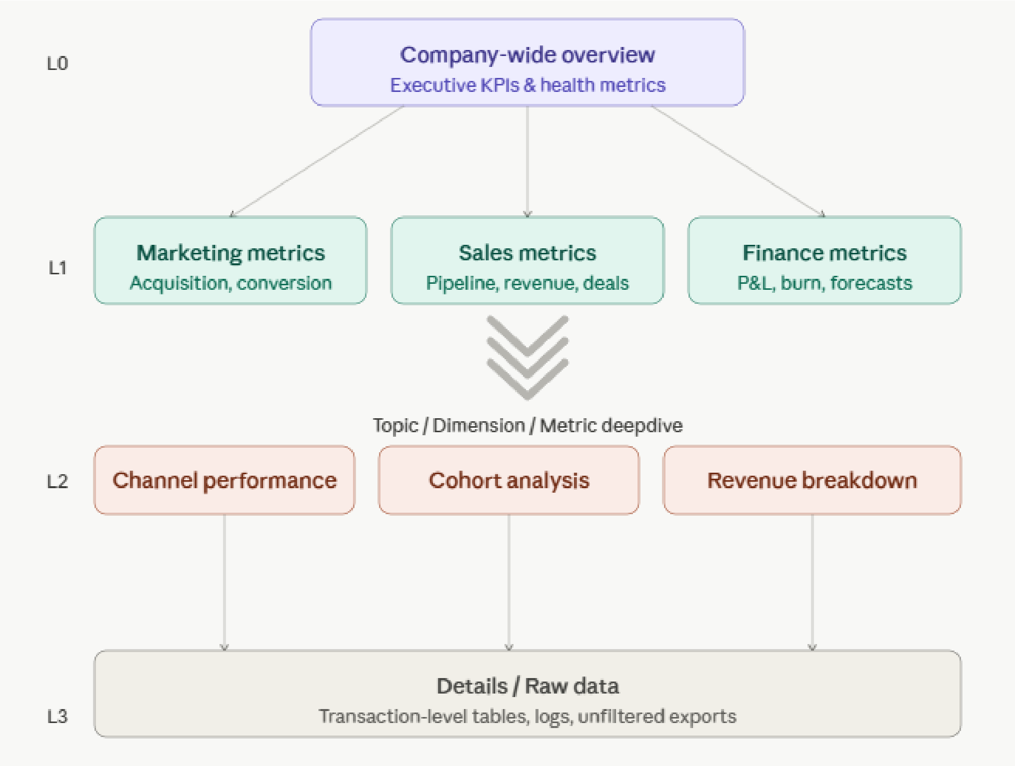
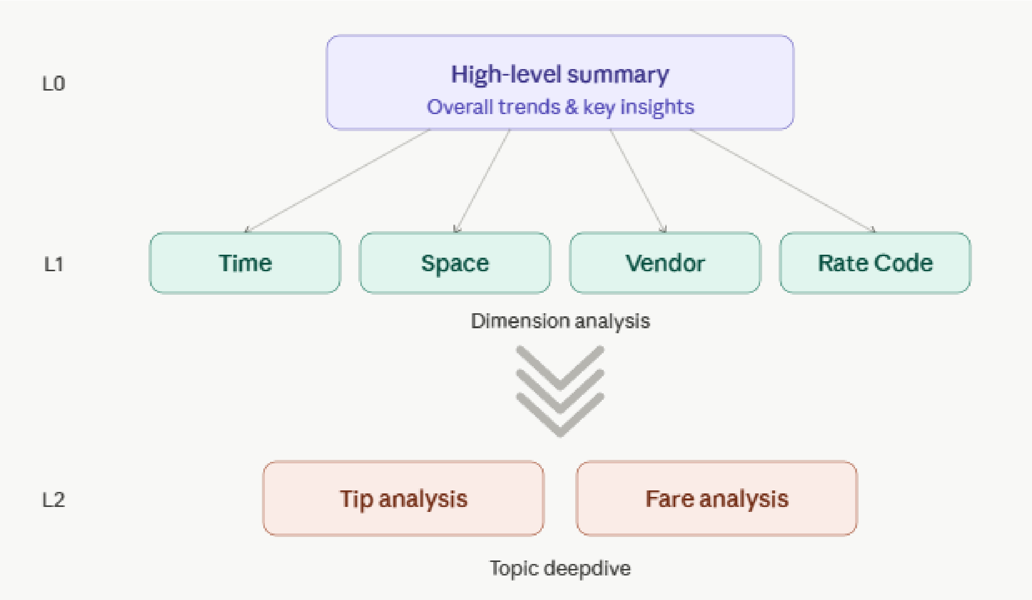
    - 5.1 L0 High-level overview
    - 5.2 L1 Time
    - 5.3 L1 Space
    - 5.4 L1 Vendor
    - 5.5 L1 Rate code
    - 5.6 L2 Tip
    - 5.7 L2 Fare
6. Conclusions



5. DATA ANALYSIS





On the left is my general data analysis framework. In this assignment, I used the modified framework (on the right)

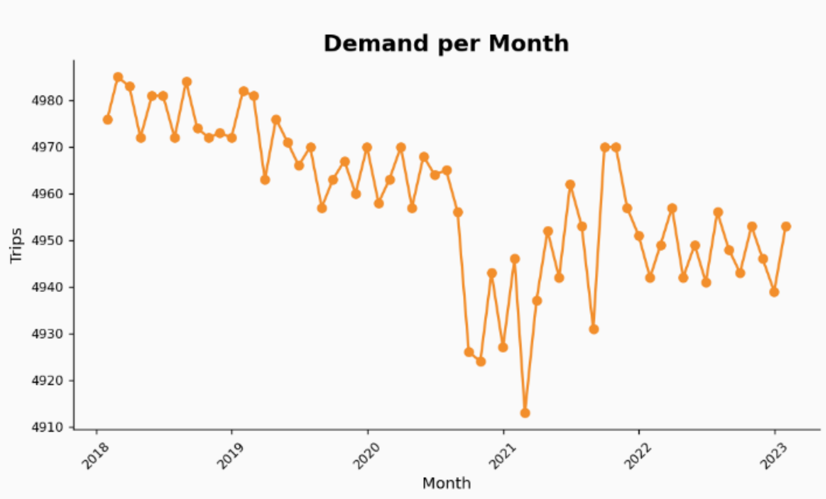
5. DATA ANALYSIS

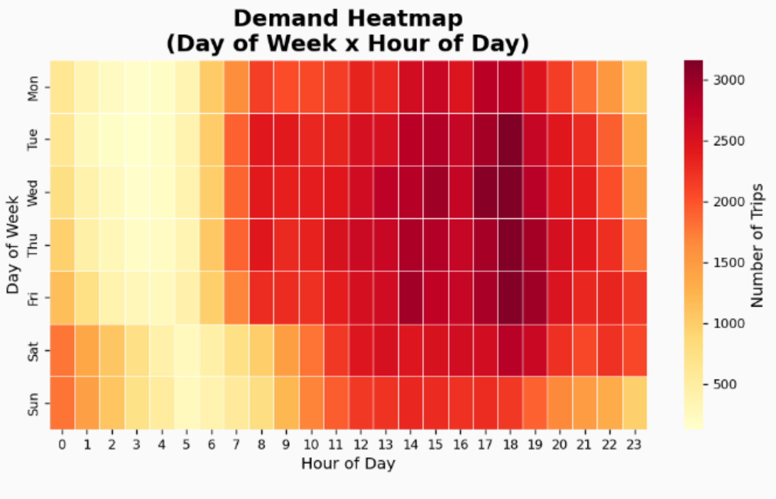
5.1 L0 High-level Overview

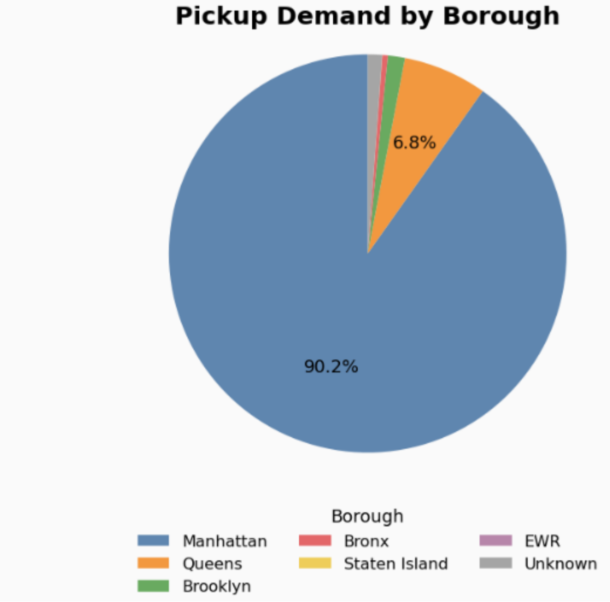
a) Demand

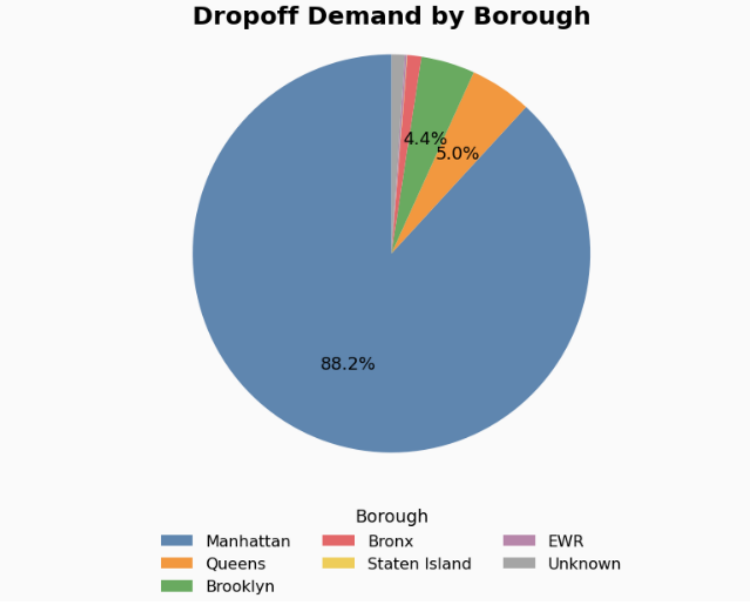
The hourly profile indicates clear late-afternoon and early-evening intensity, while day-of-week results show stronger demand on weekdays than weekends.

Demand is the highest in Manhattan (about 90% of total number of trips).







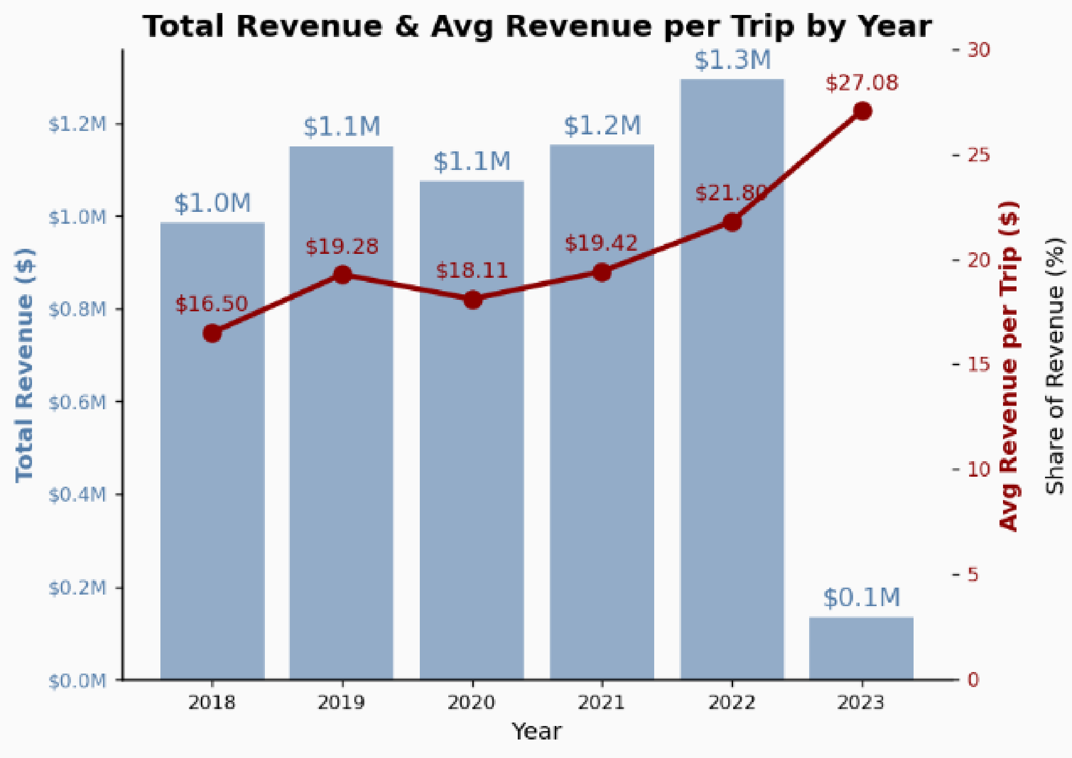


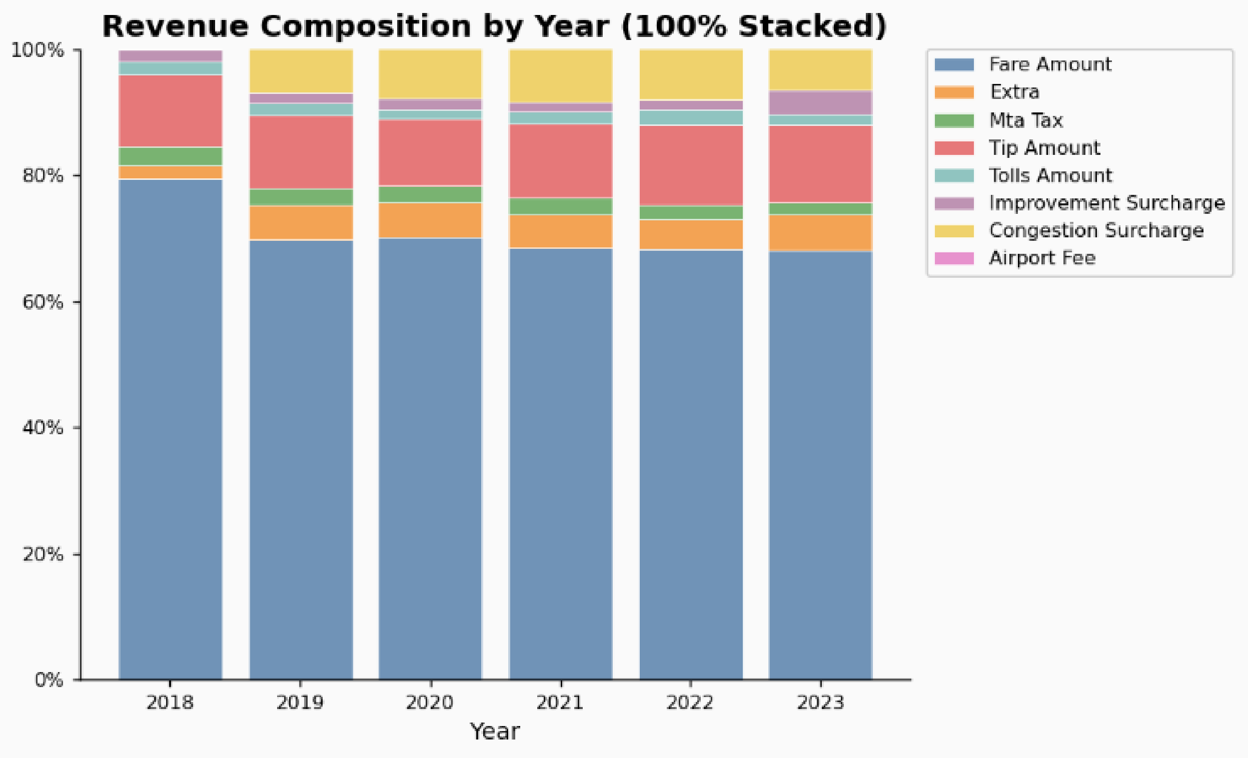
5. DATA ANALYSIS

5.1 L0 High-level Overview

b) Revenue

- Average revenue per trip trends upward and peaks in 2023.
- Revenue composition remains fare-led, with tip amount as the second-largest contributor.
- Congestion surcharge seems to be introduced in 2019.



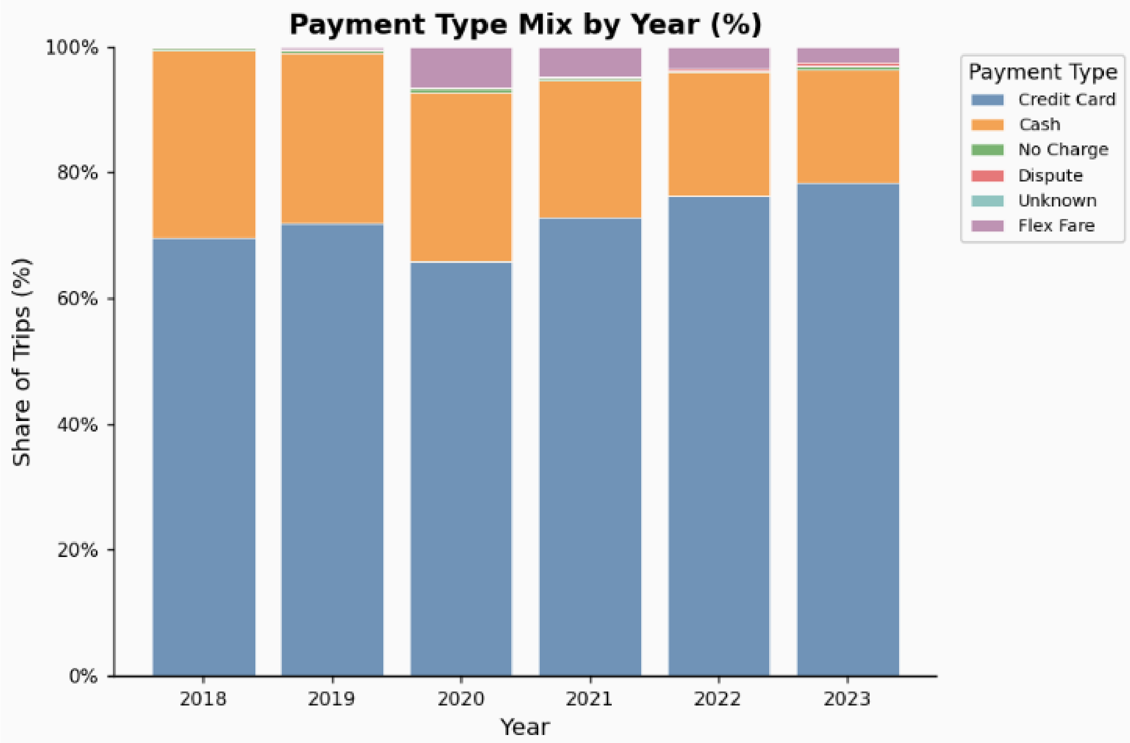


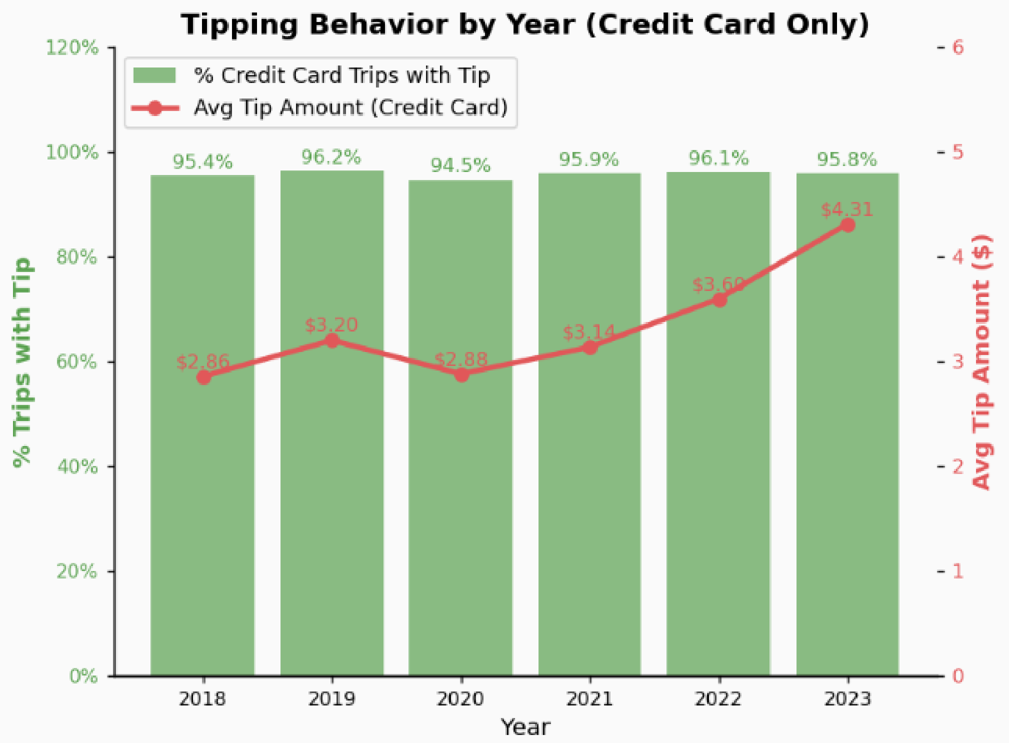
5. DATA ANALYSIS

5.1 L0 High-level Overview

c) Payment & Tip

- Card payments dominate across all years and increase over time, while cash share generally declines.
- The share of tipped trips remained stable at around 95%.
- Average tip amount also trends upward (roughly $2.9 to $4.3), with a dip in 2020 and strong growth afterward.
- Overall, payment behavior is becoming more cash-less and tip amount is improving over time.





5. DATA ANALYSIS

5.1 L0 High-level Overview

d) Efficiency

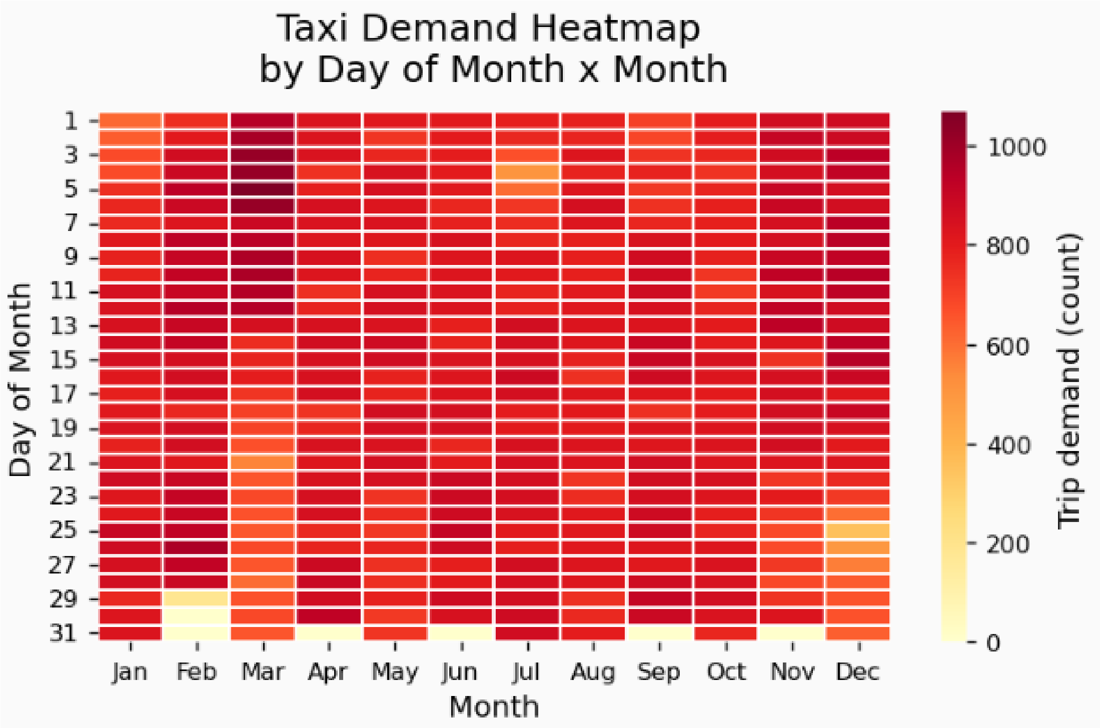
- Average trip distance stays fairly stable across years, while average trip duration slightly dipped in 2020.
- The heatmap shows clear time-dependency, with faster speeds after midnight and slower speeds during office hours on weekdays.
- Revenue per mile and per minute jumped significantly in 2023.

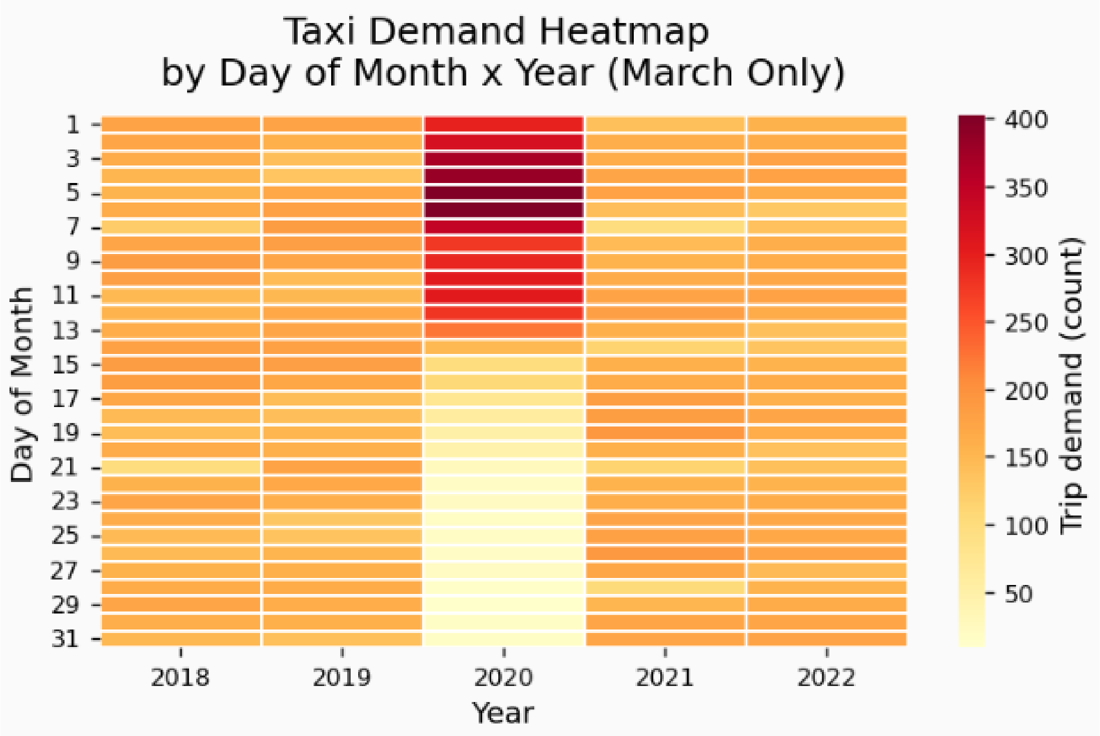
5. DATA ANALYSIS

5.2 L1 Time

- The heatmap on the left show that there is lower demand on the holidays (4 July and Christmas week). March seems highly concentrated in the early days of the month.

- The heatmap on the right explains that 2020 is the outlier, which correlates with Covid timing when there was no demand in late March 2020.





5. DATA ANALYSIS

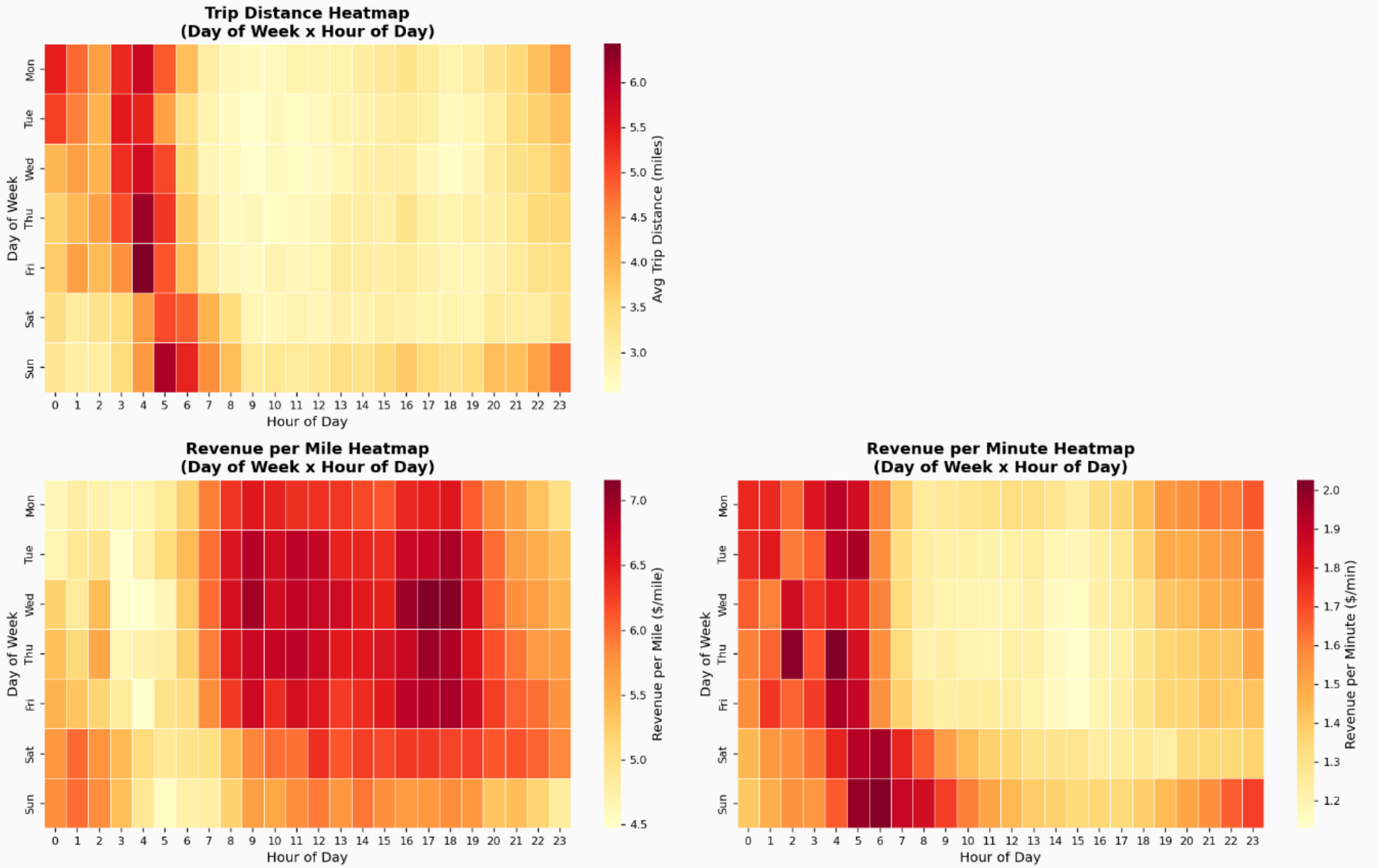
5.2 L1 Time

- Trip distance is highest in late-night to early-morning windows, especially around 3-6 AM.

- In contrast, Revenue per mile is weaker in overnight periods.

- Revenue per minute has opposite pattern to Revenue per mile.

- Together, the three heatmaps suggest distance influence revenue efficiency. Overnight periods have longer-distance trips and better Revenue per minute efficiency, but lower Revenue per mile.



5. DATA ANALYSIS

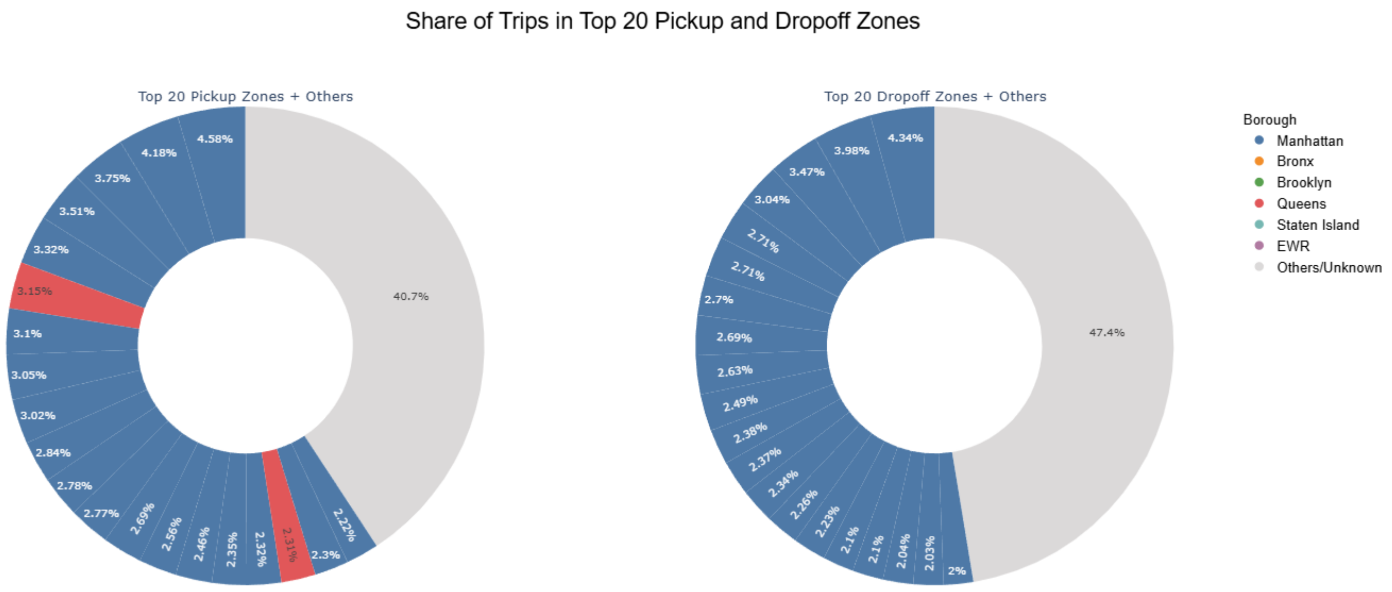
5.3 L1 Space

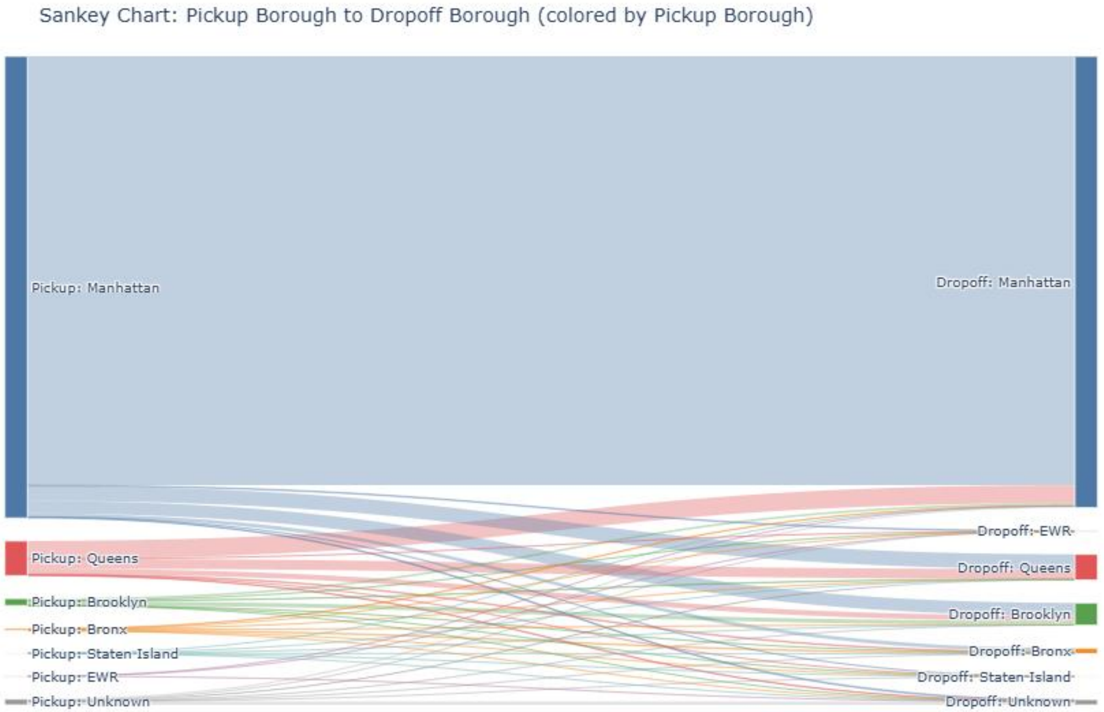
- The trips are **highly concentrated** in a few zones: out of ~250 zones, both the top 20 pickup and dropoff zones account for more than half of trip volume.

- The top 20 pickup zones include 18 in Manhattan and 2 airports in Queens.

- The top 20 dropoff zones are all in Manhattan.

- Nearly 85% of trips start and end within Manhattan, which is expected given that Manhattan is the most densely populated borough and a major commercial hub.





5. DATA ANALYSIS

5.3 L1 Space

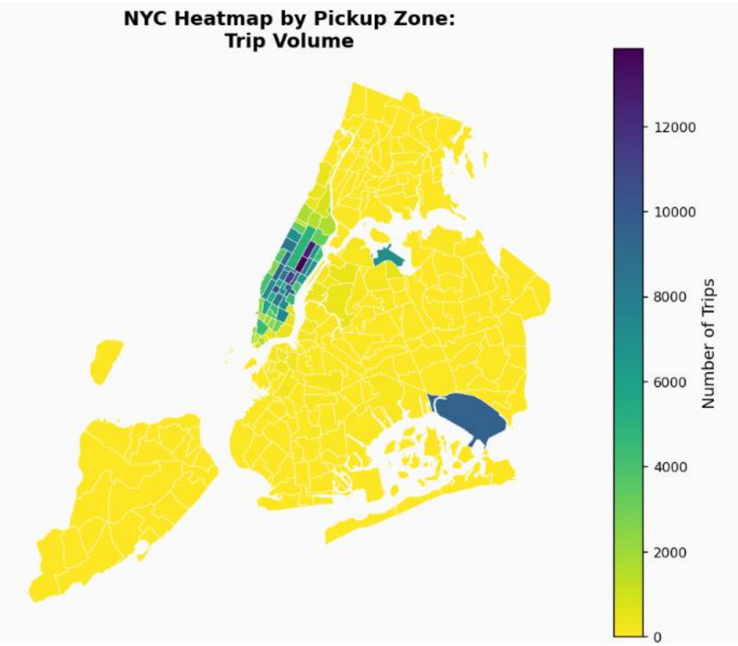
- **Trip volume** is heavily concentrated in Manhattan, with airport-related zones in Queens as secondary hotspots.

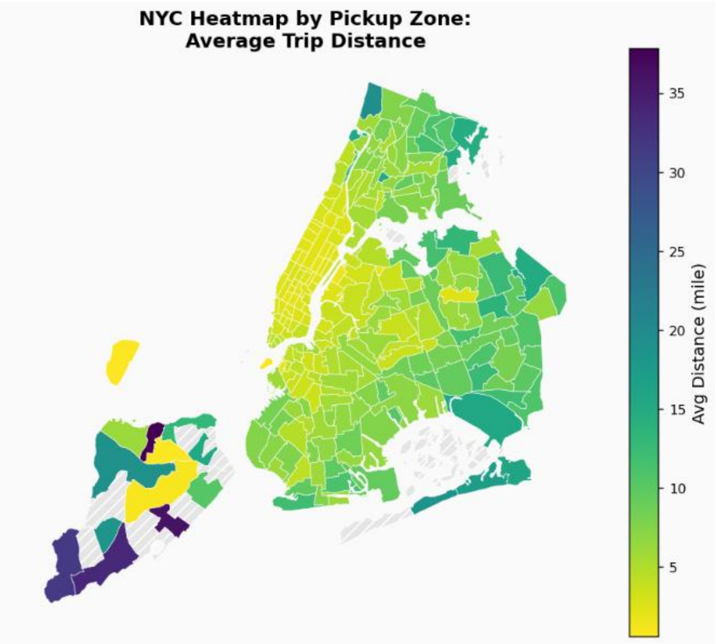
- **Average trip distance** is generally shorter in high-volume core zones (especially Manhattan) and longer in lower-volume outer-borough zones.

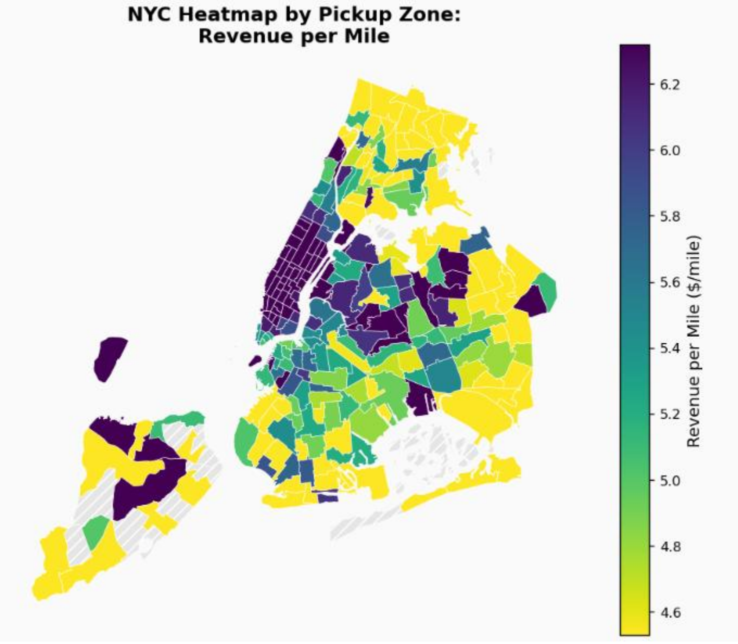
- **Revenue per mile** tends to be strongest in central, dense activity zones and weaker across many peripheral zones.

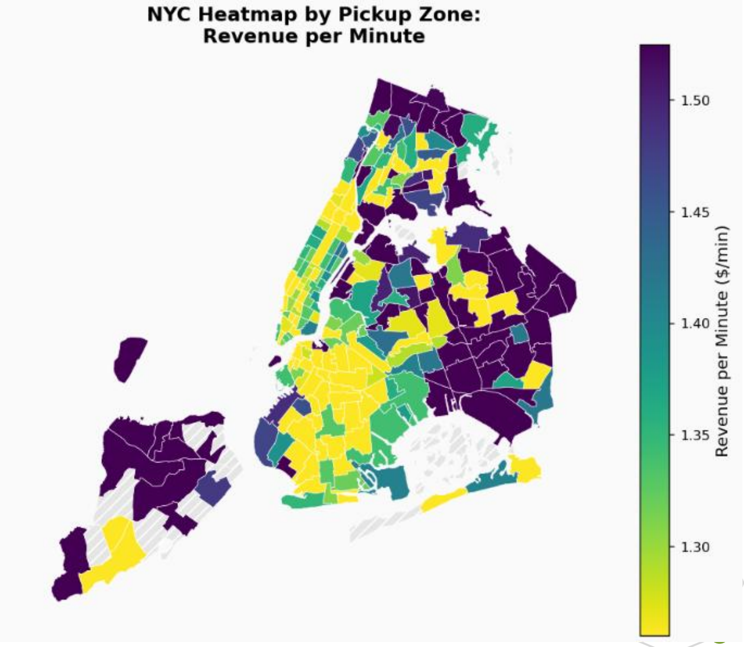
- **Revenue per minute** adds a complementary view: the general pattern is opposite to Revenue per mile.

- Together, the four maps show a clear core-periphery pattern: a high-demand urban core with stronger distance-based yield versus a lower-demand outer area with more mixed efficiency outcomes.









5. DATA ANALYSIS

5.4 L1 Vendor

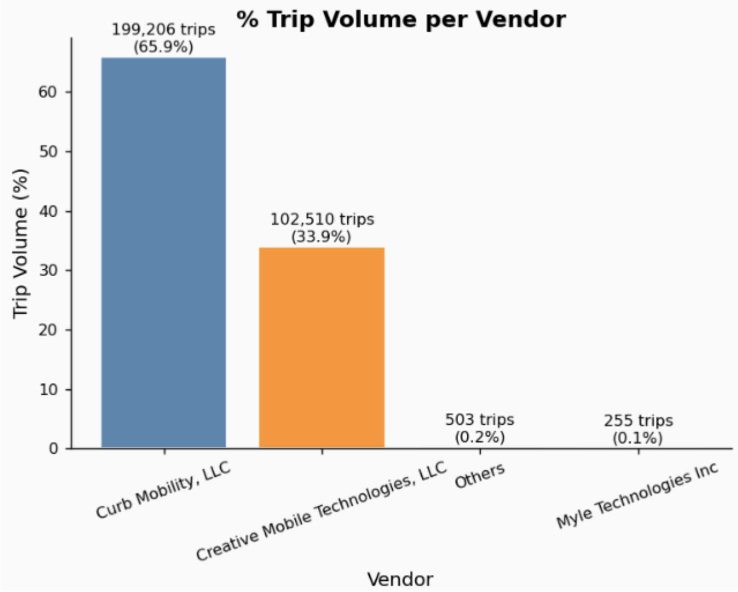
- Trip volume is highly concentrated in Curb Mobility and Creative Mobile Technologies; other vendors have minimal share.

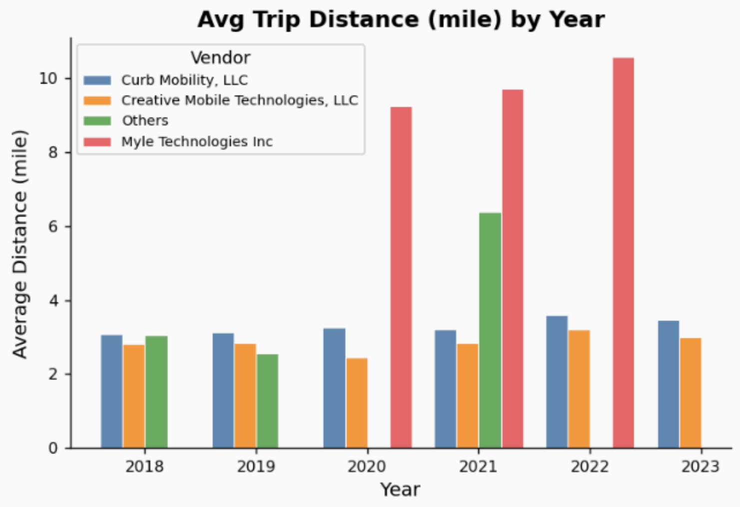
- The two major vendors maintain stable trip distance, while smaller vendors tend to run longer trips.

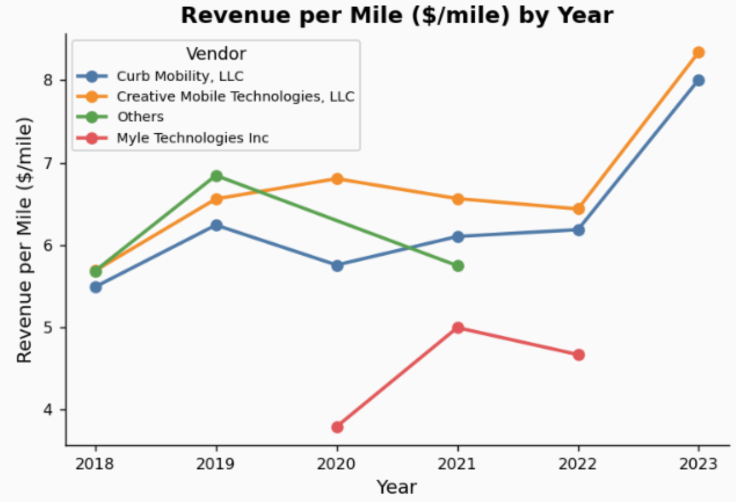
- Revenue per mile and per minute is stronger for the top vendors and jumps in 2023.

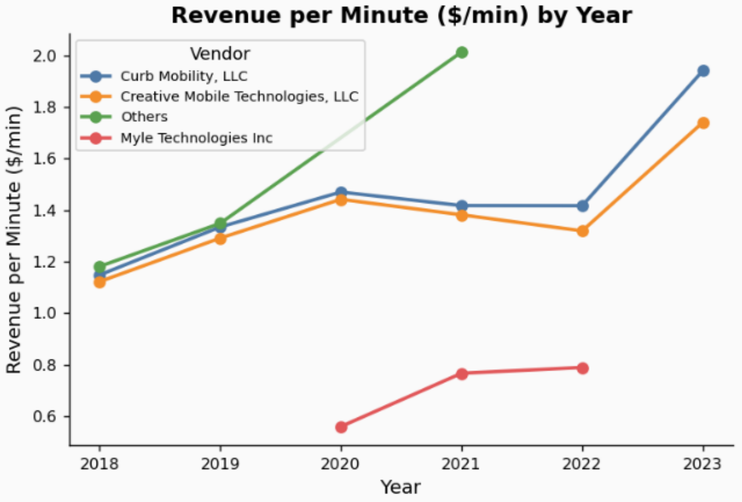
- Myle remains lower for both efficiency metrics, suggesting a different fare mix.

- Results for small vendors are volatile and should be interpreted cautiously due to low trip counts.









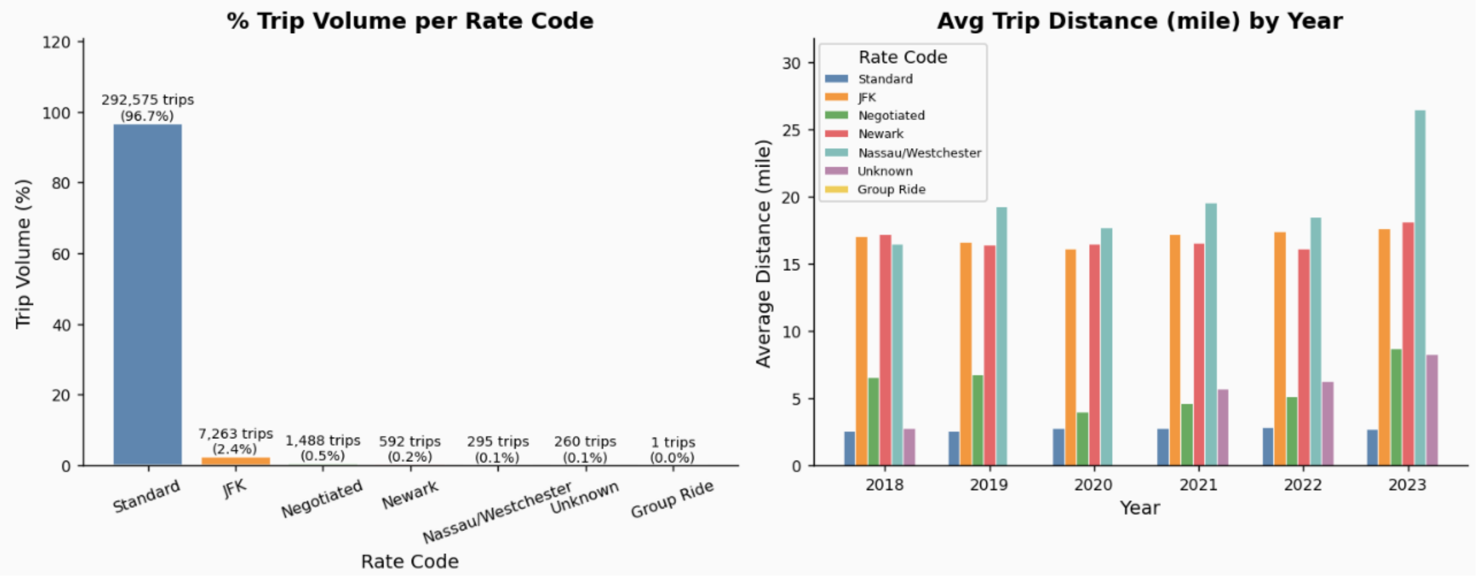
5. DATA ANALYSIS

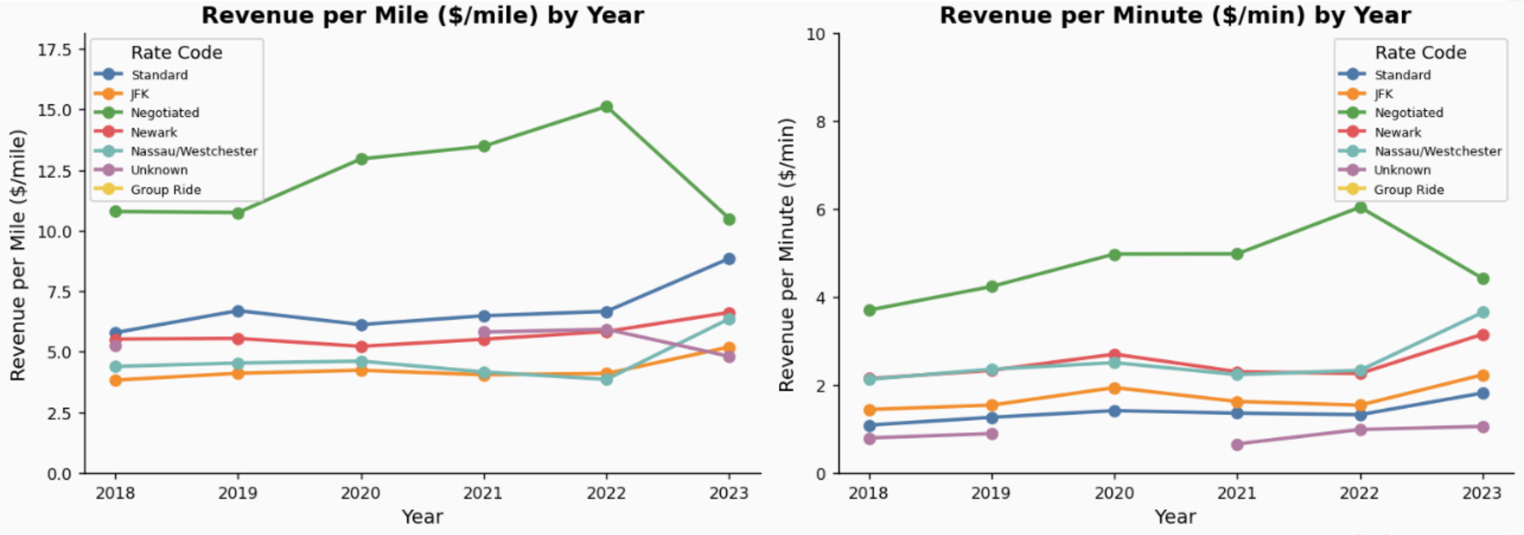
5.5 L1 Rate Code

- Standard rides drive the business by volume, while non-standard codes remain niche.

- Airport and codes carry longer trip profiles.

- Negotiated rate code perform the best for both revenue efficiency metrics.

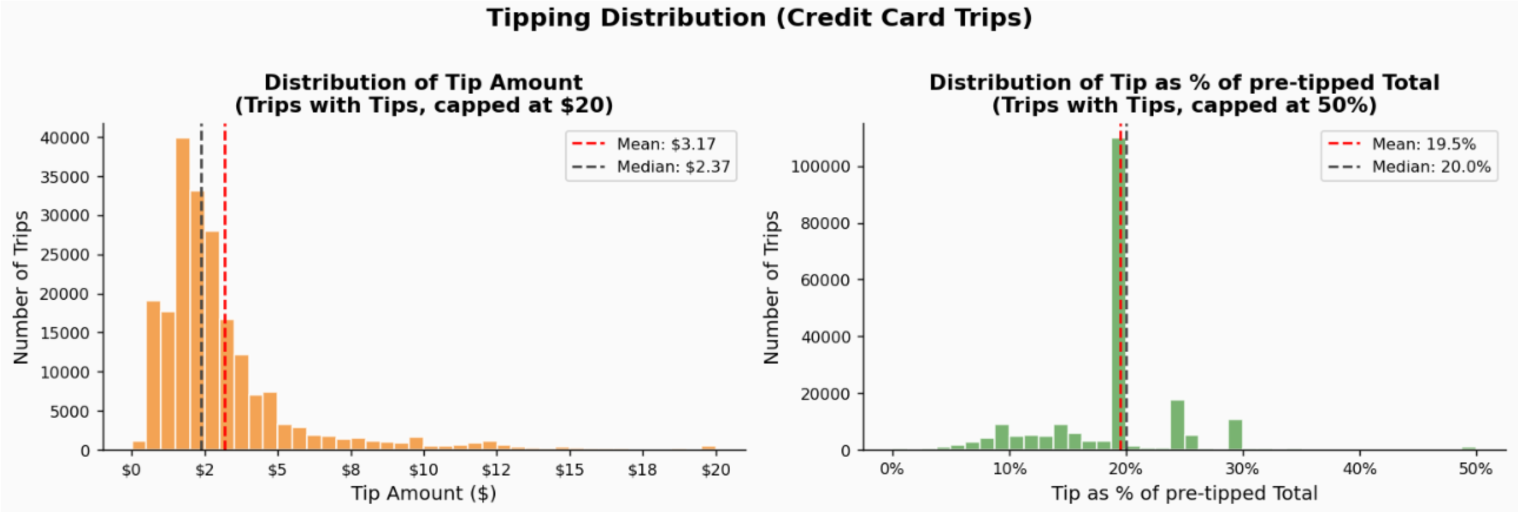


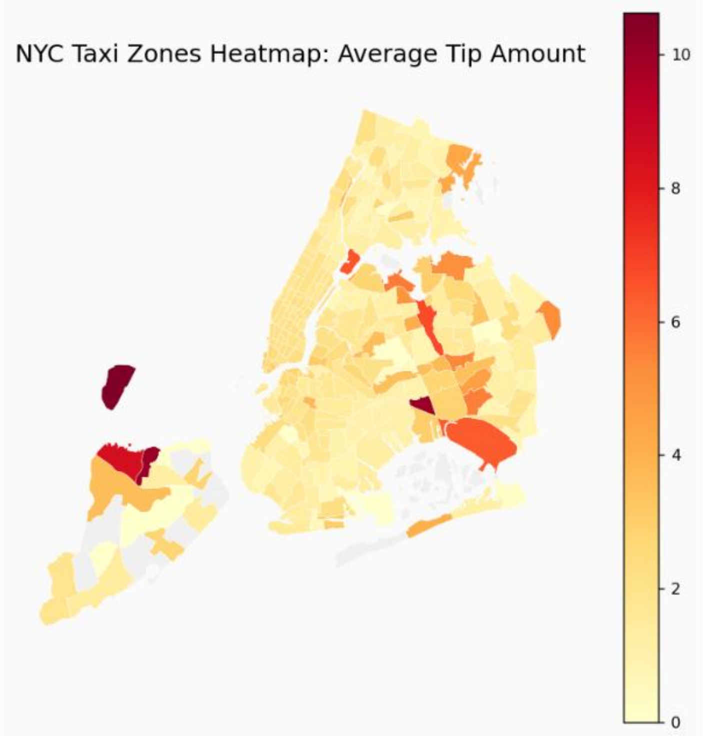


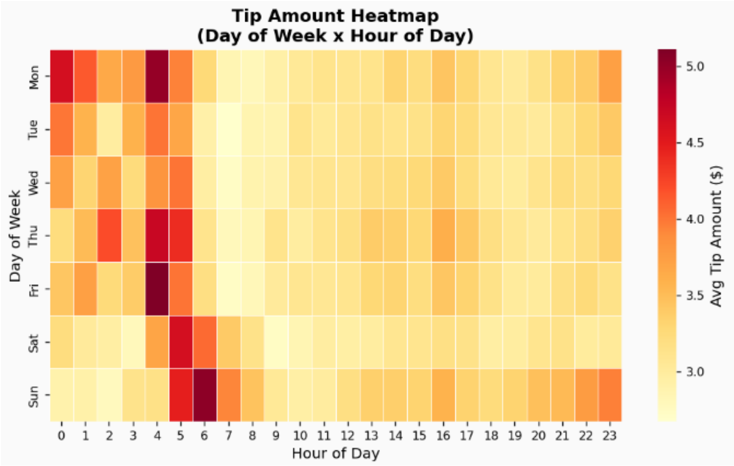
5. DATA ANALYSIS

5.6 L2 Tip

- Tips are right-skewed: most trips have low-to-moderate tips, while high tips are relatively rare.
- The concentration near small tip values suggests many short trips or low-fare rides. A thin long tail indicates occasional high-tip trips, likely linked to longer distances.
- A large majority of trips have 20% tip. This may be the default choice for customers.
- Strong positive correlation between tip amount and trip distance (Spearman test).
- Zones with the highest tips are mostly in peripheral areas and the airports.
- Tip amounts tend to be higher after midnights.



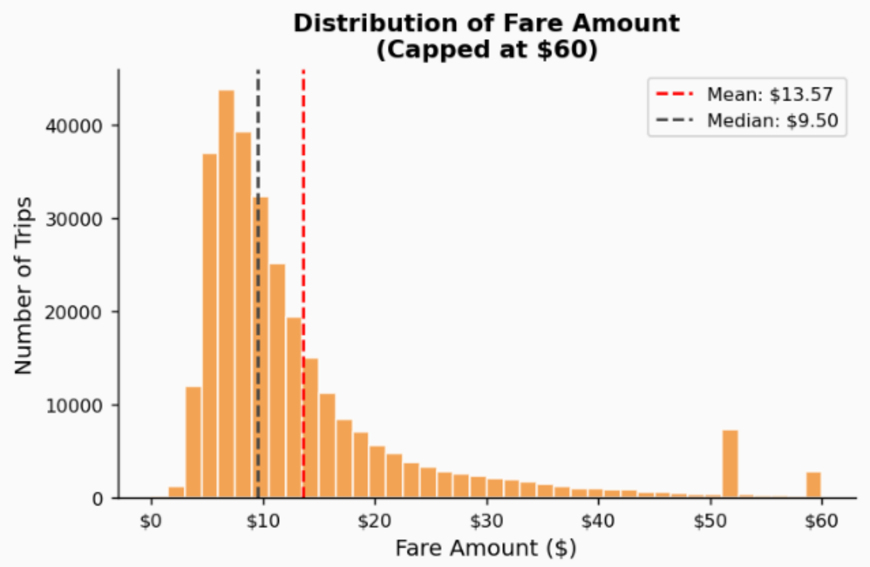


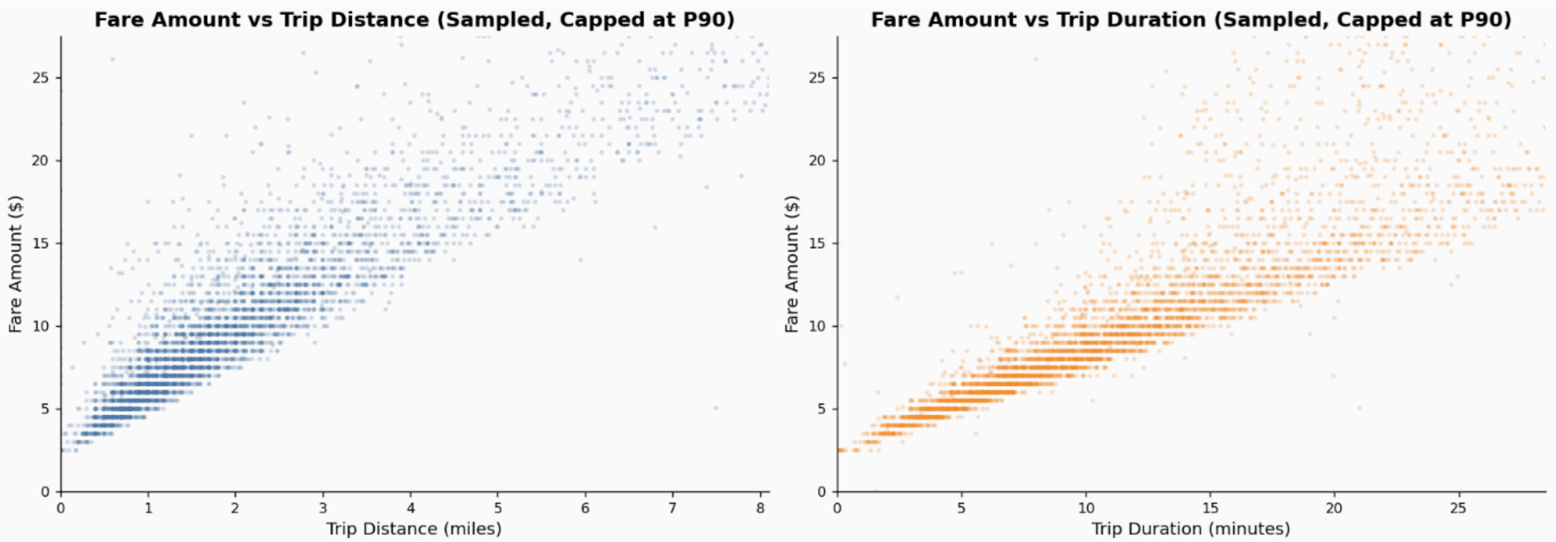


5. DATA ANALYSIS

5.7 L2 Fare

- The distribution of fare amounts is right-skewed, with a long tail of higher fares.
- Fare rises with both trip distance and trip duration, confirmed by strong positive relationships in both scatter plots and Spearman test.
- The fare-distance pattern is tighter and more linear than fare-duration, suggesting distance is the cleaner direct driver of pricing.
- The fare-duration plot is more dispersed at the top right, indicating traffic and stop-and-go conditions introduce additional fare variability beyond travel time alone.





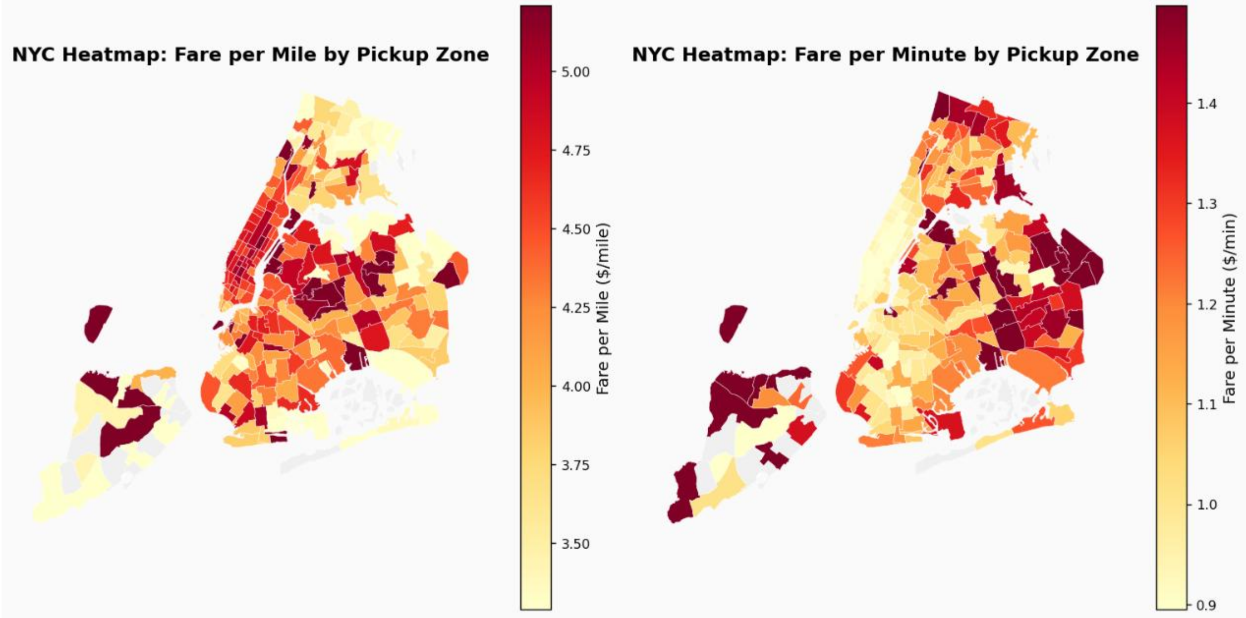
5. DATA ANALYSIS

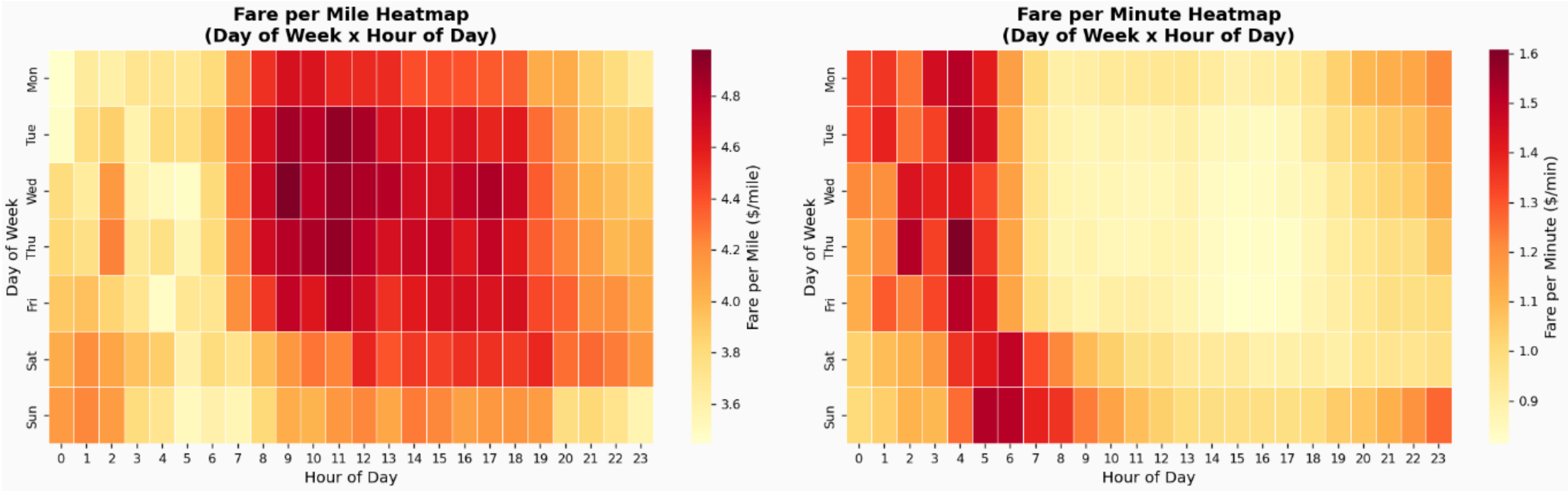
5.7 L2 Fare

- Manhattan zones generally achieve higher fare per mile but lower fare per minute, consistent with shorter, slower trips in congested urban areas.

- Temporal patterns suggest congestion increases fare per mile during peak periods, while off-peak periods tend to produce more time-efficient trips and higher fare per minute.

- The contrasting behavior of these 2 metrics supports a zone- and time-specific operating model, with both KPIs used jointly for operational monitoring and shift planning.





6. CONCLUSIONS

Key findings

- Performance is highly dependent on time and location. Citywide averages mask significant zone- and time-level differences.
- Weekdays late afternoons, and Manhattan dominates demand.
- Revenue improved over time, especially in 2023.
- Card accounts for over 75% of payment. Most trips have 20% tipping rate.
- Efficiency per mile and per minute can move in different direction, requiring management decisions to balance both.
- Demand is concentrated in two vendors and standard-rate trips.
- Fare and tips increase with trip distance and duration.

6. CONCLUSIONS

Recommendations

- Performance varies by zone and time.
- Use both $/mile and $/minute to measure efficiency.
- Demand peaks enable targeted resource allocation.
- Top-performing segments present optimization opportunities.
- Continuous data-quality monitoring is critical.

Questions?